

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Business Analysis Report

A simple business report based on the business problem, Python work, SQL analysis and Power BI dashboard.

3,900	\$59.76	3.75 / 5	27%
Customers / records	Average purchase	Average review	Subscribed

Management focus

Use customer data to improve products, customer engagement, subscriptions and marketing.

Prepared as a fresher Data Analyst portfolio project

1. Executive Summary

This project looks at 3,900 customer records to understand what customers buy, which categories bring more sales, and how customer buying habits are different. The main goal is to find simple business actions that can help improve sales, customer experience, and customer loyalty.

Clothing is the biggest category in the dashboard. It brings about \$104K in sales and has 1,737 records.

Accessories is second with about \$74K in sales and 1,240 records. Footwear and Outerwear are smaller.

Only 27% of customers are subscribers, while 73% are not. This shows that there is a large group that could be considered for a subscription offer. The company should first check which repeat buyers are most likely to join.

Young Adults are the top age group in both sales (\$62K) and customer count (1,028). The other age groups are fairly close. This means the company can use different messages for different age groups instead of using one message for everyone.

Executive takeaway

Focus first on Clothing, improve the subscription offer, and use previous purchases and buying frequency to separate new customers from repeat customers.

2. Business Problem and Objectives

The business wants to use customer shopping data to understand buying patterns and improve sales and customer engagement. The main areas are customer groups, product categories, discounts, reviews, seasons, payment methods, and repeat buying.

2.1 Business Questions

- Which customer groups bring the most sales and records?
- Do subscribers spend more than non-subscribers?
- Which products and categories are most popular or have better ratings?
- How common are discounts across products?
- Are repeat buyers more likely to subscribe?
- Does shipping type affect average purchase value?
- How can purchase frequency and previous purchases help group customers?
- Which categories and age groups should get more business attention?

2.2 Project Goals

- Clean the raw data and make it ready for Python, SQL and Power BI.
- Create a simple Power BI dashboard where users can filter key customer and sales information.
- Give recommendations that are based on the actual results from this dataset.

3. Data Overview

The dataset has 3,900 records and 18 columns before cleaning. It includes age, gender, product, category, purchase amount, location, size, color, season, rating, subscription, shipping type, discount, previous purchases, payment method, and purchase frequency.

Area	Representative fields	Business use
Customer details	Age, Gender, Location	Customer grouping and targeting
Product details	Item Purchased, Category, Size, Color	Product and demand analysis
Purchase value	Purchase Amount (USD)	Sales and purchase value analysis
Customer experience	Review Rating, Shipping Type	Customer experience check
Customer loyalty	Subscription Status, Previous Purchases, Frequency	Repeat buying analysis
Promotion / payment	Discount Applied, Payment Method	Discount and payment analysis

The business problem mentions online and offline sales, but the dataset does not have a sales channel column. So channel analysis could not be done from this data.

4. Data Preparation in Python

In Python, I first loaded the CSV file and checked the data. Then I handled missing values, cleaned the column names, created a few useful columns, removed one repeated column, and loaded the cleaned data into MySQL.

4.1 Data Checks

There were 37 missing values in Review Rating. The other columns had no missing values. The average purchase amount is \$59.76, the average review rating is 3.75, the average age is 44.07, and the average number of previous purchases is 25.35.

37	0	44.07	25.35
Missing review ratings	Missing in other fields	Average age	Average prior purchases

4.2 Changes Made to the Data

- The missing review ratings were filled using the median rating of the same category.
- Column names were changed to lowercase with underscores. Purchase amount was renamed to purchase_amount.
- An age_group column was created using age quartiles. This gave four age groups for the analysis.
- Purchase frequency was converted into approximate days. For example, Weekly = 7 days, Fortnightly / Bi-Weekly = 14 days, Monthly = 30 days, Quarterly / Every 3 Months = 90 days, and Annually = 365 days.
- promo_code_used was removed because it had the same value as discount_applied for all records.

Why this matters

The aim was to keep the data simple. Each change makes the data easier to use in SQL and Power BI and removes repeated information.

5. SQL Analysis

After cleaning the data, it was loaded into a MySQL table called customer. SQL was then used to answer the main business questions.

5.1 SQL Questions

ID	Question	Why I asked it
Q1	Revenue by gender	Compare sales by gender.
Q2	Discount + above-average spend	Find discounted purchases that are above average spend.
Q3	Top-rated products	Find the products with the best ratings.
Q4	Shipping comparison	Compare average spend for Standard and Express shipping.
Q5	Subscription behavior	Compare subscribers and non-subscribers.
Q6	Discount-heavy products	Find products that use discounts most often.
Q7	Customer lifecycle	Group customers as New, Returning or Loyal.
Q8	Top items within category	Find the top three products in each category.
Q9	Repeat buyers and subscription	Check whether repeat buyers are

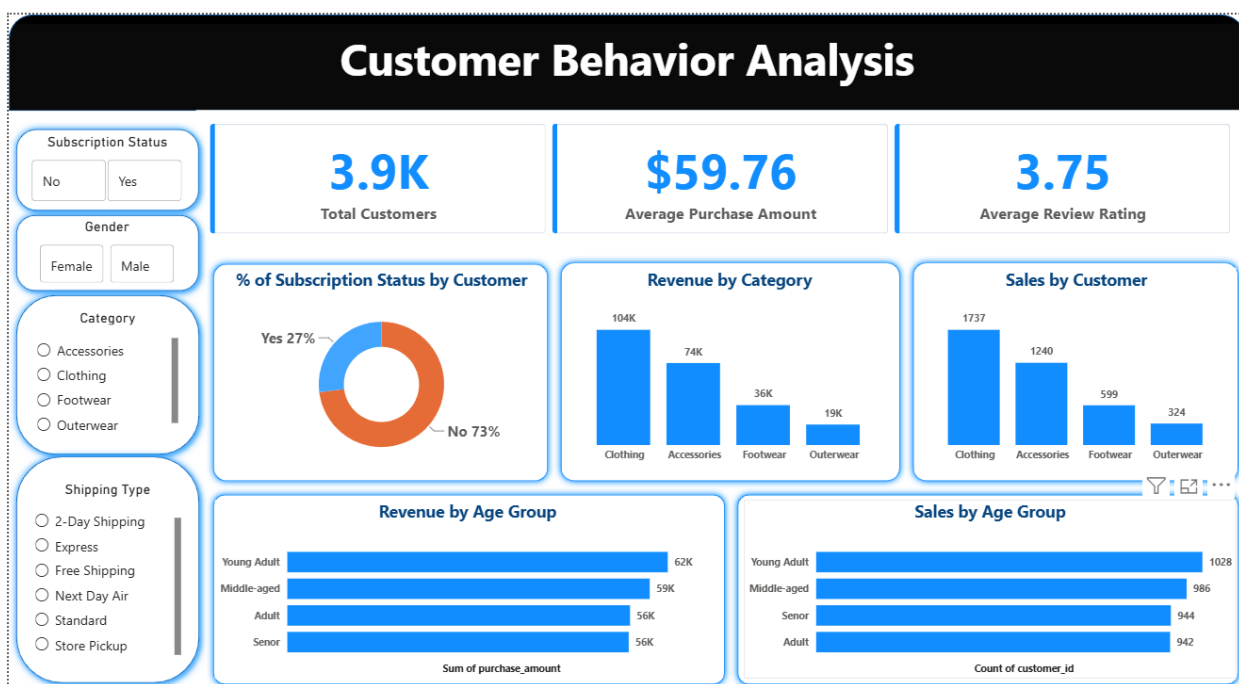
ID	Question	Why I asked it
		more likely to subscribe.
Q10	Revenue by age group	Compare sales across age groups.

The SQL part shows how business questions can be answered with group by, filters, subqueries, CASE statements, and window functions. This also makes the analysis easier to repeat.

6. Power BI Dashboard

The Power BI dashboard brings the main results into one page. It has filters for subscription status, gender, category, and shipping type. The main KPIs are total customers, average purchase amount, and average review rating.

Figure 1. Power BI dashboard used for this project.



Customers	Average purchase	Average rating	Subscribers
3,900	\$59.76	3.75 / 5	27%

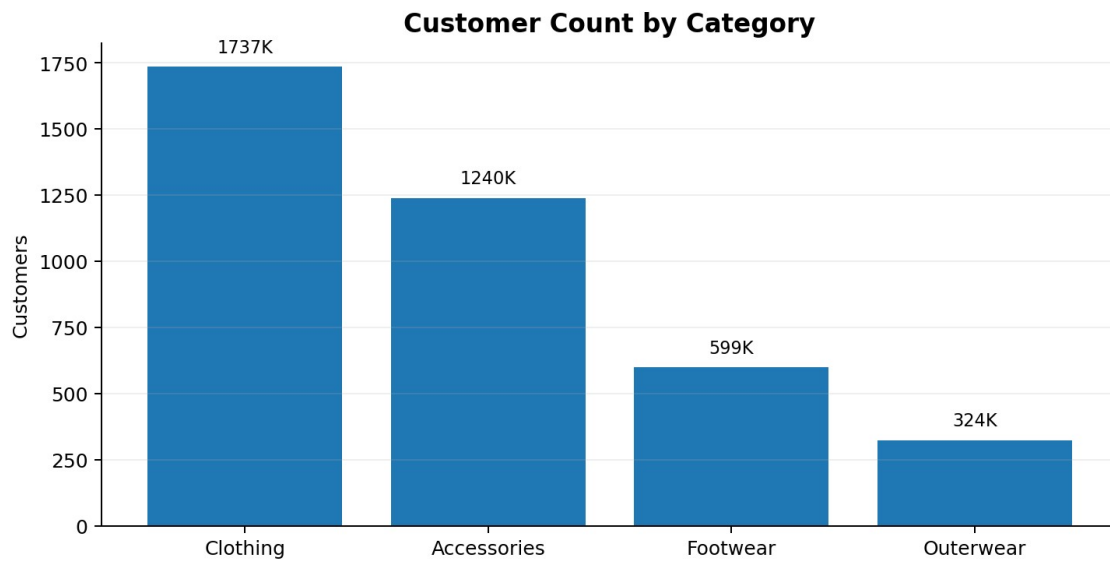
7. Key Findings

7.1 Category Results

Clothing is the top category with about \$104K in sales and 1,737 records. Accessories is second with about \$74K and 1,240 records. Together, these two categories make about \$178K of the roughly \$233K shown in the category chart.

Figure 2. Sales by category from the dashboard.

Figure 3. Customer count by category from the dashboard.



7.2 Age Group Results

Young Adults are the top age group with about \$62K in sales and 1,028 records. Middle-aged customers are next with about \$59K and 986 records. Senior and Adult groups are also close. The results are fairly balanced across the age groups.

Figure 4. Sales by age group.

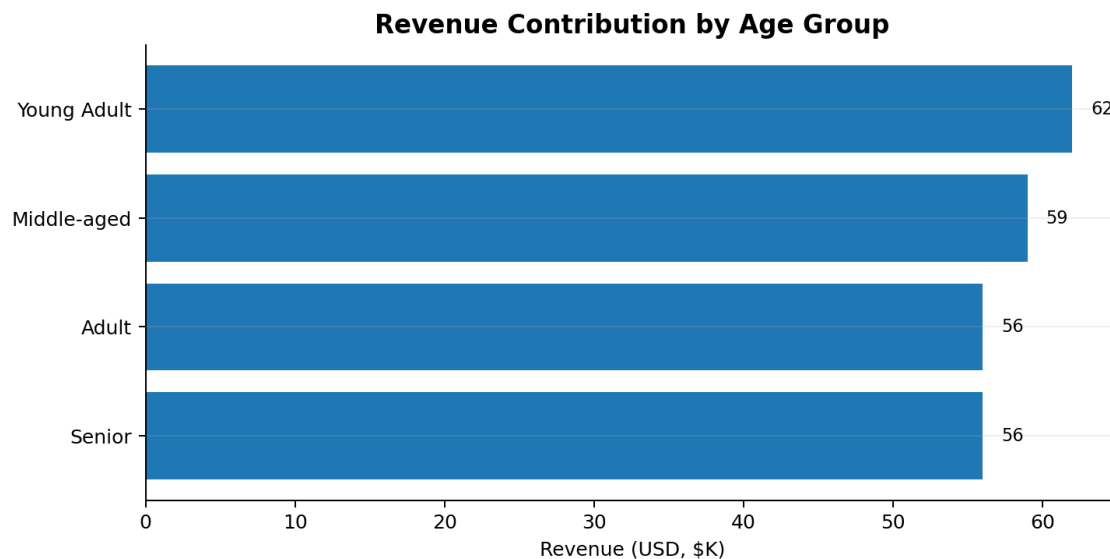
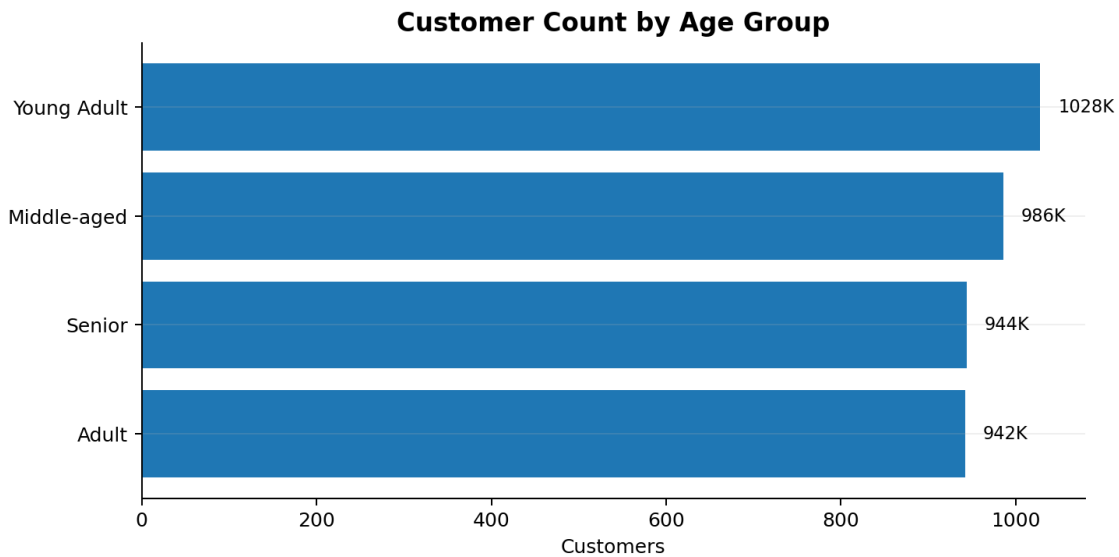


Figure 5. Customer count by age group.



7.3 Subscription Result

Only 27% of customers are subscribers and 73% are not. The SQL analysis compares average spend and total revenue for both groups. A good next step is to check whether subscribers also have better repeat buying before offering bigger subscription benefits.

8. Business Recommendations

01 | Focus on the top category

Use Clothing as the main category for product planning, bundles and marketing because it has the highest sales and activity.

02 | Improve the subscription offer

Target non-subscribers with benefits based on how often they buy instead of giving the same offer to everyone.

03 | Use different messages for different customers

Use previous purchases and buying frequency to create different messages for new, returning and loyal customers.

04 | Use discounts carefully

Use the SQL discount results to find products that may depend too much on discounts. Avoid large discounts on products that already sell well.

05 | Watch customer ratings

The average rating is 3.75, so reviews should be tracked. Low-rated categories can be checked for product or service problems.

06 | Add more data for future analysis

Add sales channel and transaction date if the business wants online/offline analysis and time trends later.

9. Limitations and Next Steps

- The dataset does not have a transaction date, so month-to-month and year-to-year trends cannot be measured.
- The business problem mentions online and offline sales, but there is no channel column in the dataset.
- The dashboard shows summary values. Before using the results for real business decisions, the main numbers should be checked again from the cleaned SQL table.
- The current dataset is better for customer profile and behavior analysis. It is not detailed enough for full order-level cohort or customer lifetime value analysis.
- The notebook contains a local database password. It is not included in this report and should not be uploaded to GitHub.

Next step
Add transaction date, sales channel, order ID and more customer history. This will allow deeper analysis such as customer repeat buying, customer lifetime value and campaign results.

10. Conclusion

This project shows a full data analysis process: understand the business problem, clean the data in Python, analyze it in SQL, and present the results in Power BI. The main results show that Clothing and Accessories are the biggest categories, most customers are not subscribers, and buying behavior is different across age groups and previous purchases.

The main point of the project is how the tools work together. Python prepares the data, SQL answers business questions, and Power BI makes the results easy to understand. The recommendations are then linked to the findings.

Portfolio tip
Explain the project as a simple business story: What was the problem? How did I prepare the data? What did I find? What should the business do next?

Appendix A. Source Files

This report is based on the supplied Business Problem Document, Python notebook, SQL file, and Power BI dashboard screenshot. The wording and layout were changed, but the main analysis and reported numbers were kept from the source files.

Source file	Used for
Business Problem Document.pdf	Business context, main business question and project deliverables
customer_behavior.ipynb	Dataset structure, data checks, changes made in Python and MySQL loading
customer_behavior_sql.sql	Ten business questions and SQL queries

Source file	Used for
Power BI dashboard screenshot	Main numbers and category / age-group dashboard findings